panel_economic_caract

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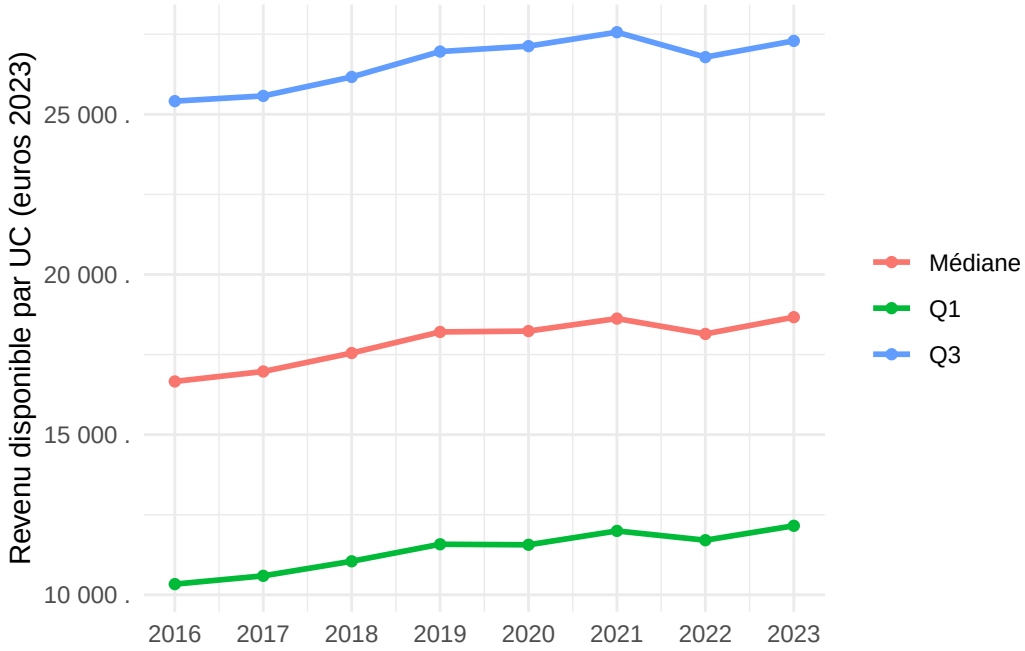
1 Goal

The goal of this script is to begin linking panel data with idealista prices




```
Key: <q2016>
   q2016     1     2     3     4     5
   <int> <num> <num> <num> <num> <num>
1:     1  48.2  26.8  14.6   7.2   3.3
2:     2  22.3  37.8  23.6  12.1   4.3
3:     3   8.8  25.7  34.4  22.8   8.2
4:     4   4.0   9.2  25.8  40.3  20.7
5:     5   2.0   3.0   6.5  22.3  66.2
```

```
    year pct_pers_hh_vide pct_pers_hh_incomplet pers_hh_extreme
   <int>            <num>                <num>           <int>
1:  2016              1.7                 18.0            2676
2:  2017              0.6                  9.3            2464
3:  2018              0.6                  9.6            3044
4:  2019              0.7                 10.3            2982
5:  2020              0.8                 10.6            2580
6:  2021              0.5                  8.2            2328
7:  2022              0.6                  8.7            2544
8:  2023              0.6                  9.0            2314
```

```
    year ecart_moyen_sources
   <int>               <num>
```

1:	2016	0.01
2:	2017	0.01
3:	2018	0.01
4:	2019	0.00
5:	2020	0.01
6:	2021	0.01
7:	2022	0.00
8:	2023	0.00

1.1 Income distribution

	year	var	moyenne	p10	p25	p50	p75	p90	p90_p10	gini	top10	top1
	<int>	<char>	<num>	<num>	<num>	<num>	<num>	<num>	<num>	<num>	<num>	<num>
1:	2016	eq_rb	24009	5713	10804	18454	30027	45789	8.01	0.4273	31.03	7.87
2:	2017	eq_rb	24434	6069	11204	18914	30356	46069	7.59	0.4226	30.87	7.93
3:	2018	eq_rb	25228	6463	11737	19609	31118	46948	7.26	0.4191	30.83	8.16
4:	2019	eq_rb	26127	6960	12345	20381	32175	48375	6.95	0.4138	30.56	8.08
5:	2020	eq_rb	26167	6936	12286	20393	32443	48827	7.04	0.4133	30.36	7.81
6:	2021	eq_rb	26960	7426	12837	20991	33199	49896	6.72	0.4102	30.41	7.85
7:	2022	eq_rb	26385	7206	12581	20585	32414	48613	6.75	0.4118	30.55	8.06
8:	2023	eq_rb	27003	7593	13097	21229	33170	49668	6.54	0.4059	30.14	7.77
9:	2016	eq_rbd	20125	5447	10334	16660	25408	36664	6.73	0.3890	28.07	6.74
10:	2017	eq_rbd	20402	5802	10591	16970	25571	36737	6.33	0.3844	27.91	6.78
11:	2018	eq_rbd	21044	6179	11046	17547	26164	37397	6.05	0.3804	27.88	7.01
12:	2019	eq_rbd	21743	6639	11578	18205	26955	38365	5.78	0.3744	27.61	
13:	2020	eq_rbd	21772	6627	11561	18232	27124	38671	5.84	0.3731	27.37	
14:	2021	eq_rbd	22276	7127	11994	18622	27558	39241	5.51	0.3692	27.37	
15:	2022	eq_rbd	21677	6890	11704	18143	26781	38050	5.52	0.3708	27.50	
16:	2023	eq_rbd	22105	7257	12155	18666	27290	38643	5.32	0.3642	27.08	

```
12:  6.93
13:  6.66
14:  6.68
15:  6.86
16:  6.58
```

Key: <year>

year	eq_rb	eq_rbd	effet_redistributif
<int>	<num>	<num>	<num>
2016	0.4273	0.3890	0.0383
2017	0.4226	0.3844	0.0382
2018	0.4191	0.3804	0.0387
2019	0.4138	0.3744	0.0394
2020	0.4133	0.3731	0.0402
2021	0.4102	0.3692	0.0410
2022	0.4118	0.3708	0.0410
2023	0.4059	0.3642	0.0417

Key: <dec_rb>

dec_rb	2016	2017	2018	2019	2020	2021	2022	2023
<int>	<num>	<num>	<num>	<num>	<num>	<num>	<num>	<num>
1	6.8	7.2	7.6	7.2	7.0	7.3	8.1	9.4
2	4.3	5.4	4.6	4.8	4.6	4.4	4.7	4.7
3	5.1	5.2	5.4	5.5	5.4	6.0	6.4	6.5
4	6.4	6.9	7.1	7.3	7.1	7.7	8.3	8.5
5	8.5	8.9	9.1	9.3	9.2	10.0	10.6	10.7
6	10.7	11.2	11.4	11.7	11.5	12.2	12.7	13.1
7	13.0	13.4	13.6	13.9	13.9	14.6	15.0	15.4
8	15.5	15.9	16.1	16.4	16.4	17.0	17.4	17.9
9	18.5	18.8	18.9	19.2	19.3	19.8	20.3	20.8
10	24.5	24.8	24.9	25.2	25.3	25.9	26.4	26.8

year	seuil	taux
<int>	<num>	<num>
2016	9996.0	23.7
2017	10182.0	23.0
2018	10528.2	22.5
2019	10923.0	22.0
2020	10939.2	22.3
2021	11173.2	22.1
2022	10885.8	22.0
2023	11199.6	21.6

Key: <age_band>

age_band	2016	2017	2018	2019	2020	2021	2022	2023
<fctr>	<num>	<num>	<num>	<num>	<num>	<num>	<num>	<num>
<18	31.8	30.7	30.3	29.7	30.1	29.2	29.4	28.8
18-34	28.4	26.9	26.4	25.8	26.3	24.9	24.8	24.0
35-49	25.1	23.7	23.1	22.4	22.8	21.6	21.7	21.4
50-64	20.9	20.7	20.3	19.8	20.3	19.7	19.8	19.3
65-74	12.2	12.6	12.7	13.0	13.1	14.9	14.7	14.6
75+	12.7	13.8	13.8	13.8	13.3	17.8	17.3	17.9

Key: <sex>

sex	2016	2017	2018	2019	2020	2021	2022	2023
<char>	<num>	<num>	<num>	<num>	<num>	<num>	<num>	<num>
	NA	NA	NA	NA	48.1	NA	NA	NA
1	23.8	22.8	22.3	21.8	22.1	21.5	21.4	21.1
6	23.7	23.1	22.8	22.3	22.5	22.7	22.6	22.1

1.2 Sources of income

1.2.1 Distribution of source of income by decile

	year	dec_rbd	eq_wages	eq_pens	eq_ben	eq_cap	eq_rent	eq_bus	eq_gains
	<int>	<int>	<num>	<num>	<num>	<num>	<num>	<num>	<num>
1:	2016	1	50.3	13.2	36.4	1.5	1.9	-4.3	1.1
2:	2016	2	50.2	22.9	17.1	0.7	1.0	7.5	0.6
3:	2016	3	51.9	30.2	9.8	0.7	1.0	6.0	0.5
4:	2016	4	55.7	28.4	7.6	0.8	1.3	5.7	0.5
5:	2016	5	58.5	27.5	5.7	0.9	1.6	5.3	0.5
6:	2016	6	61.6	25.6	4.4	1.0	1.8	5.0	0.6
7:	2016	7	63.7	24.7	3.5	1.0	2.1	4.5	0.5
8:	2016	8	65.9	22.6	2.7	1.2	2.5	4.3	0.7
9:	2016	9	68.1	19.8	2.1	1.5	3.2	4.5	0.9
10:	2016	10	61.8	13.9	1.3	5.2	5.4	7.8	4.6
11:	2023	1	49.4	14.4	31.2	0.8	1.8	2.8	-0.4
12:	2023	2	52.6	21.3	17.5	0.4	1.2	6.2	0.9
13:	2023	3	54.6	27.2	11.0	0.4	1.2	4.8	0.7
14:	2023	4	60.4	24.3	8.3	0.4	1.5	4.4	0.7
15:	2023	5	63.1	23.8	6.3	0.4	1.6	4.1	0.6
16:	2023	6	64.8	23.1	5.0	0.5	1.9	4.0	0.7
17:	2023	7	65.6	22.5	4.0	0.6	2.4	4.1	0.7
18:	2023	8	67.0	20.6	3.4	0.8	3.0	4.4	0.9
19:	2023	9	65.9	20.1	2.7	1.1	3.9	5.0	1.3

```
20:  2023      10     57.7    12.5    1.7    5.9     6.2    9.6      6.4
     year dec_rbd eq_wages eq_pens eq_ben eq_cap eq_rent eq_bus eq_gains
    <int>   <int>    <num>   <num>  <num>  <num>   <num>  <num>    <num>
    eq_other
       <num>
 1:        0
 2:        0
 3:        0
 4:        0
 5:        0
 6:        0
 7:        0
 8:        0
 9:        0
10:        0
11:        0
12:        0
13:        0
14:        0
15:        0
16:        0
17:        0
18:        0
19:        0
20:        0
    eq_other
       <num>
```

Variable	Source	Definition
`eq_wages`	M91	Wages, employer pension contributions, severance pay
`eq_pens`	M921 + R1212	Pensions, incl. exempt incapacity pensions
`eq_ben`	M922 + M923 − R1212	Unemployment, family, dependency and other benefits, incl. IMV
`eq_cap`	M93	Interest, dividends and other capital income
`eq_rent`	M94	Rental income from real estate

Variable	Source	Definition
eq_bus	M95	Self-employment and business income
eq_gains	M96	Net capital gains
eq_other	M97	Other income under special tax regimes

All `eq_*` variables: household total ÷ modified OECD scale, in 2023 euros. The eight sources sum to `eq_rb`.

1.2.2 Rental income

```
    year pct_hh_loyer part_loyers_top10
   <int>        <num>              <num>
1:  2016         13.8               52.6
2:  2017         14.6               52.5
3:  2018         15.1               52.9
4:  2019         15.4               52.6
5:  2020         15.6               50.7
6:  2021         15.9               51.1
7:  2022         16.2               52.1
8:  2023         16.5               52.2
```

- ~ 15% of people perceive rental income
- The proportion of people collecting rental income increases every year between 2016 and 2023
- People in the highest decile of disposable income collect half of the total rents

1.2.3 Unequal distribution of the income source

```
Key: <source>
     source    2016    2017    2018    2019    2020    2021    2022    2023
     <char>   <num>   <num>   <num>   <num>   <num>   <num>   <num>   <num>
1:   eq_ben -0.6346 -0.6798 -0.2718 -0.2832 -0.5426 -0.7407 -0.5822 -0.6661
2:   eq_bus  6.7174  7.0773  7.2904  7.1945  6.9700  8.1710  8.3867  8.3368
3:   eq_cap  4.1344  4.0534  4.3047  4.5212  3.7989  3.5266  4.0815  4.5585
4: eq_gains  3.4735  4.3768  4.8689  4.7854  4.3716  5.9578  5.9761  4.9550
5: eq_other  0.0052  0.0040  0.0044  0.0022  0.0045  0.0007 -0.0071 -0.0187
6:  eq_pens 13.7500 13.3758 12.9298 12.7104 13.8322 12.4299 12.6710 12.3021
```

```
7:  eq_rent  4.7066  4.9369  5.1580  5.3044  5.0076  5.1201  5.3342  5.5570
8: eq_wages 67.8475 66.8556 65.7156 65.7651 66.5577 65.5346 64.1397 64.9756
```

	year	source	part	concentration	contribution	gini	pct_contribution
	<int>	<char>	<num>	<num>	<num>	<num>	<num>
1:	2016	eq_wages	0.6198	0.4678	0.2899	0.4273	67.8475
2:	2016	eq_pens	0.2086	0.2817	0.0588	0.4273	13.7500
3:	2016	eq_ben	0.0403	-0.0673	-0.0027	0.4273	-0.6346
4:	2016	eq_cap	0.0238	0.7436	0.0177	0.4273	4.1344
5:	2016	eq_rent	0.0315	0.6375	0.0201	0.4273	4.7066
6:	2016	eq_bus	0.0574	0.5003	0.0287	0.4273	6.7174
7:	2016	eq_gains	0.0186	0.7959	0.0148	0.4273	3.4735
8:	2016	eq_other	0.0000	0.7904	0.0000	0.4273	0.0052
9:	2017	eq_wages	0.6200	0.4556	0.2825	0.4226	66.8556
10:	2017	eq_pens	0.2051	0.2756	0.0565	0.4226	13.3758
11:	2017	eq_ben	0.0386	-0.0743	-0.0029	0.4226	-0.6798
12:	2017	eq_cap	0.0219	0.7811	0.0171	0.4226	4.0534
13:	2017	eq_rent	0.0329	0.6349	0.0209	0.4226	4.9369
14:	2017	eq_bus	0.0587	0.5097	0.0299	0.4226	7.0773
15:	2017	eq_gains	0.0228	0.8116	0.0185	0.4226	4.3768
16:	2017	eq_other	0.0000	1.0370	0.0000	0.4226	0.0040
17:	2018	eq_wages	0.6183	0.4455	0.2754	0.4191	65.7156
18:	2018	eq_pens	0.1988	0.2726	0.0542	0.4191	12.9298
19:	2018	eq_ben	0.0425	-0.0268	-0.0011	0.4191	-0.2718
20:	2018	eq_cap	0.0223	0.8083	0.0180	0.4191	4.3047
21:	2018	eq_rent	0.0339	0.6374	0.0216	0.4191	5.1580
22:	2018	eq_bus	0.0595	0.5138	0.0306	0.4191	7.2904
23:	2018	eq_gains	0.0247	0.8266	0.0204	0.4191	4.8689
24:	2018	eq_other	0.0000	1.1351	0.0000	0.4191	0.0044
25:	2019	eq_wages	0.6202	0.4388	0.2721	0.4138	65.7651
26:	2019	eq_pens	0.1973	0.2666	0.0526	0.4138	12.7104
27:	2019	eq_ben	0.0436	-0.0269	-0.0012	0.4138	-0.2832
28:	2019	eq_cap	0.0225	0.8331	0.0187	0.4138	4.5212
29:	2019	eq_rent	0.0346	0.6335	0.0220	0.4138	5.3044
30:	2019	eq_bus	0.0579	0.5137	0.0298	0.4138	7.1945
31:	2019	eq_gains	0.0239	0.8302	0.0198	0.4138	4.7854
32:	2019	eq_other	0.0000	0.6985	0.0000	0.4138	0.0022
33:	2020	eq_wages	0.5967	0.4610	0.2751	0.4133	66.5577
34:	2020	eq_pens	0.2034	0.2810	0.0572	0.4133	13.8322
35:	2020	eq_ben	0.0754	-0.0297	-0.0022	0.4133	-0.5426
36:	2020	eq_cap	0.0186	0.8451	0.0157	0.4133	3.7989
37:	2020	eq_rent	0.0335	0.6170	0.0207	0.4133	5.0076

	year	source	part	concentration	contribution	gini	pct_contribution
	<int>	<char>	<num>	<num>	<num>	<num>	<num>
38:	2020	eq_bus	0.0499	0.5767	0.0288	0.4133	6.9700
39:	2020	eq_gains	0.0224	0.8050	0.0181	0.4133	4.3716
40:	2020	eq_other	0.0000	0.8115	0.0000	0.4133	0.0045
41:	2021	eq_wages	0.6048	0.4445	0.2688	0.4102	65.5346
42:	2021	eq_pens	0.1947	0.2619	0.0510	0.4102	12.4299
43:	2021	eq_ben	0.0592	-0.0513	-0.0030	0.4102	-0.7407
44:	2021	eq_cap	0.0173	0.8364	0.0145	0.4102	3.5266
45:	2021	eq_rent	0.0337	0.6236	0.0210	0.4102	5.1201
46:	2021	eq_bus	0.0603	0.5562	0.0335	0.4102	8.1710
47:	2021	eq_gains	0.0301	0.8119	0.0244	0.4102	5.9578
48:	2021	eq_other	0.0000	0.5841	0.0000	0.4102	0.0007
49:	2022	eq_wages	0.6128	0.4311	0.2642	0.4118	64.1397
50:	2022	eq_pens	0.1943	0.2686	0.0522	0.4118	12.6710
51:	2022	eq_ben	0.0479	-0.0500	-0.0024	0.4118	-0.5822
52:	2022	eq_cap	0.0198	0.8493	0.0168	0.4118	4.0815
53:	2022	eq_rent	0.0349	0.6293	0.0220	0.4118	5.3342
54:	2022	eq_bus	0.0615	0.5619	0.0345	0.4118	8.3867
55:	2022	eq_gains	0.0288	0.8534	0.0246	0.4118	5.9761
56:	2022	eq_other	0.0000	0.9811	0.0000	0.4118	-0.0071
57:	2023	eq_wages	0.6167	0.4276	0.2637	0.4059	64.9756
58:	2023	eq_pens	0.1919	0.2602	0.0499	0.4059	12.3021
59:	2023	eq_ben	0.0476	-0.0568	-0.0027	0.4059	-0.6661
60:	2023	eq_cap	0.0224	0.8271	0.0185	0.4059	4.5585
61:	2023	eq_rent	0.0359	0.6286	0.0226	0.4059	5.5570
62:	2023	eq_bus	0.0605	0.5592	0.0338	0.4059	8.3368
63:	2023	eq_gains	0.0252	0.7987	0.0201	0.4059	4.9550
64:	2023	eq_other	-0.0001	1.0210	-0.0001	0.4059	-0.0187

`part`	mean(source) / mean(`eq_rb`)	Weight of the source in total income. Example: 0.60 = 60% of gross income
`concentration`	`conc(source, rank in eq_rb)`	Where the source sits in the distribution. Close to 1: concentrated among the rich. Close to 0: spread evenly. Negative: concentrated among the poor
`contribution`	`part × concentration`	Gini points contributed by the source. Contributions sum to the Gini

`gini`	Gini of `eq_rb`	Total Gini, same on every row of a given year
`pct_contribution`	$100 \times$ `contribution` / `gini`	Percentage of total inequality due to the source
`residu_arrondi`	`gini` $-\ \Sigma$ `contribution`	Should be close to 0. Non-zero when sources do not add up exactly to `eq_rb` (rounding, missing components)